One-site Hubbard model with PIKs

Kenji Fukushima,[1] Friederike Ihssen,[1] Renzo Kapust,[2] and Jan M. Pawlowski[2,3]

[1] *Theoretische Physik III, Ruhr-Universität Bochum, Universitätsstraße 150, 44721 Bochum, Germany*
[2] *Institut für Theoretische Physik, Universität Heidelberg, Philosophenweg 16, 69120 Heidelberg, Germany*
[3] *ExtreMe Matter Institute EMMI, GSI, Planckstr. 1, 64291 Darmstadt, Germany*

I. INTRODUCTION

Main conceptual insights of the paper

- Poles and ripping of the real PIKfold
- Classification of sign-problem "classes" and complex PIKfolds
- Weighted sums of PIKfolds

- Friederike: Fill up the practical part as much as possible
- Renzo: Other (numerical) implementation
- Jan: Collecting conceptual part & two stage general process
- Kenji: Thimbles et al, **??**

Next chat:
[**JMP:** taken from the complex PIK paper [**?**] for completeness]

II. THE ONE-SITE HUBBARD MODEL

A. Model definition and analytical solution

The hubbard model suffers the sign problem in the same way as QCD at finite density. Since it describes the lattice fermions, the model has a conserved charge corresponding to the fermion number, which can be coupled to the chemical potential μ. In the tight-binding limit, fermions can no longer hop to neighboring sites, and the model Hamiltonian is decomposed into the one-site Hamiltonian at each site. More specifically, using the number operator for up and down spin fermions, $\hat{n}_\uparrow$ and $\hat{n}_\downarrow$, the one-site Hamiltonian reads:

$$\hat{H} = U\hat{n}_\uparrow\hat{n}_\downarrow - \mu(\hat{n}_\uparrow + \hat{n}_\downarrow)\,. \tag{1}$$

This Hamiltonian is simple enough to be analytically solvable, and yet, it still captures the essence of the sign problem.

In the operator formalism, it is a straightforward task to compute the partition function using the Fock state basis. Because $\hat{n}_{\uparrow,\downarrow}$ takes an eigenvalue of either 0 or 1, there are four eigenvalues of $\hat{H}$, namely, 0, $-\mu$, $-\mu$, $U-2\mu$. Therefore, the partition function is

$$Z = \mathrm{tr}\big[e^{-\beta\hat{H}}\big] = 1 + 2\,e^{\beta\mu} + e^{\beta(2\mu-U)}\,, \tag{2}$$

from which we can compute thermodynamic quantities.

The equivalent representation in the path-integral formalism is

$$Z = \mathcal{N}\int d\phi\left(1+e^{\beta(i\phi+\mu+U/2)}\right)^2 e^{\beta\phi^2/2U} \tag{3}$$

with the normalization, $\mathcal{N}=\sqrt{\beta/(2\pi U)}$. The part of the integrand in the parentheses is a counterpart of the Dirac determinant in QCD at finite density. We emphasize that the similarity is attributed to the fermionic nature and in this sense the one-site Hubbard model can be regarded as a well-motivated toy model to investigate the finite-QCD sign problem.

Indeed, if we consider the strong-coupling limit of QCD in the hopping parameter expansion, the leading-order Dirac determinant reads:

$$\sim \prod_{\text{phase space}} \det\big(1+L\,e^{\beta\mu}\big)\big(1+L^\dagger\,e^{-\beta\mu}\big)\,, \tag{4}$$

where L is a $SU(3)$ element corresponding to $e^{\beta i\phi}\in U(1)$ in the one-site Hubbard model. We observe that, in the same way as the tight-binding Hubbard model, the spatial hopping is suppressed, and the above determinant is evaluated at each phase-space point. Then, it is easy to see the origin of the sign problem. For simplicity, let us keep terms up to the linear order in terms of L, and then, we can expand the determinant as

$$\sim 1 + \mathrm{tr}\,L\,e^{\beta\mu} + \mathrm{tr}\,L^\dagger\,e^{-\beta\mu}\,. \tag{5}$$

at each phase-space point. Obviously, for $\mu\neq 0$, the Hermiticity is broken, and the eigenvalue takes a complex value including a negative real part.

Coming back to Eq. (**??**), it is useful to expand the integrand as $Z=Z_0+Z_1+Z_2$, where

$$Z_0 = \mathcal{N}\int d\phi\, e^{\beta\phi^2/2U}\,, \tag{6a}$$

$$Z_1 = 2\mathcal{N}\int d\phi\, e^{\beta\phi^2/2U+i\beta\phi+\beta(\mu+U/2)}\,, \tag{6b}$$

$$Z_2 = \mathcal{N}\int d\phi\, e^{\beta\phi^2/2U+2i\beta\phi+2\beta(\mu+U/2)}\,. \tag{6c}$$

These are Gaussian integrals individually. The first integral of Z_0 is 1 with a proper choice of the normalization $\mathcal{N}$. The next one, Z_1, looks complex, but noting that $\beta\phi^2/2U+i\beta\phi+\beta(\mu+U/2)=\beta(\phi+iU)^2/2U+\beta\mu$, we can make a variable shift as $\phi+iU\to\phi$ to find that the Gaussian integral leads to $2e^{\beta\mu}$, that is nothing but the

second term in Eq. (**??**). We can shift $\phi + 2iU \to \phi$ for Z_2 to reproduce the third term in Eq. (**??**).

One might think that the one-site Hubbard model is too simple, but once the fermionic determinant is included in the action, we see that the total action is a good proxy for the QCD sign problem, that is,

$$S(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{\beta(i\phi+\mu+U/2)}\right] . \tag{7}$$

We point out that this action has some characteristic points. First, we can easily identify the zeros of the partition function where the action is singular. These singular points are aligned along

$$\phi_0 = (2n+1)\frac{\pi}{\beta} + i\left(\mu + \frac{U}{2}\right) . \tag{8}$$

Second, we can find the stationary points (sometimes called the critical points) from the condition, $S'(\phi) = 0$, where $S'(\phi) = dS(\phi)/d\phi$.

B. Structures of the Lefschetz Thimbles

III. PIKS AND THE SIGN PROBLEM

How are PIKs being used to compute in the presence of a sign-problem

IV. APPLICATIONS

A. Paths and poles

In the presence of poles in the action, the choice of the path becomes increasingly important, as it determines if the PIKfold at a given time-slice corresponds to a valid integration contour or not.

To demonstrate the impact of poles on the PIKfold we begin by considering a simple path which evolve samples from a Gaussian distribution at $t = 0$ to the full action **??** at $t = 1$

$$S_t(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{t\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right] . \tag{9a}$$

with time dependent poles

$$\phi_0 = (2n+1)\frac{\pi t}{\beta} + i\left(\mu + \frac{U}{2}\right) . \tag{9b}$$

Throughout the RG-time evolution of the distribution, the poles remain at a constant imaginary value and only move in the direction of the real axis. Hence, any PIKfold that remains below $\mathrm{Im}[\mathcal{M}_t] \lesssim \mu + \frac{U}{2}$ does not encounter the poles.

We depict the final sampling manifold for this example in **??** for the parameter choices $U = 1$, $\beta = 10$ and various densities. The initial PIKfold is initiated on the real axis $\mathcal{M}_0 = \mathbb{R}$. The results displayed in figures (a) and (b) correspond to $\mu = 1.75$ and $\mu = 1.65$ and are valid sampling manifolds as they remain far away enough from the poles and show no ripping. In (c) at $\mu = 1.5$, the poles lie on the sampling manifold. As such it does no longer form a closed integration contour and we can no longer evaluate observables from the samples.

Importantly, a slightly different path choice corresponds to a different time-dependent location of the poles. For example

$$S_t(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + te^{\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right] . \tag{10a}$$

has time dependent poles

$$\phi_0 = (2n+1)\frac{\pi}{\beta} + i\left(\mu + \frac{U}{2} + \frac{\log(t)}{\beta}\right) . \tag{10b}$$

that move very strongly in the imaginary direction, starting at $-\infty$ at $t = 0$ and moving up to their final location at $t = 1$.

For the parameter choices above, we find that $\mu = 1.75$ does not yield a closed contour at $t = 1$, which is displayed in **??**. This is explained by the poles pushing though the PIKfold at some finite t.

However, we also find that the path choice **??** performs worse than **??** for $\mu \sim -0.5$. In these examples the initial manifold $\mathcal{M}_0 = \mathbb{R}$ lies on top of the poles **??** of the first example, whereas the dynamics of the second path is more controlled. This is shown in **??** and **??** at $\mu = -0.675$ for both paths.

For values of $\mu < -0.7$ we find that both paths result in closed manifolds that are fit for sampling.

The conclusion is that depending on the scenario at hand the different time-dependence of the pole structure may help: It is important, that the PIKfold does not cross the poles. The fact that there are poles somewhere in the complex plane is not a problem as long as they dont intersect the manifold throughout the time evolution. This also highlights the very "local" structure of the PIKs.

B. Complex initial sampling manifolds

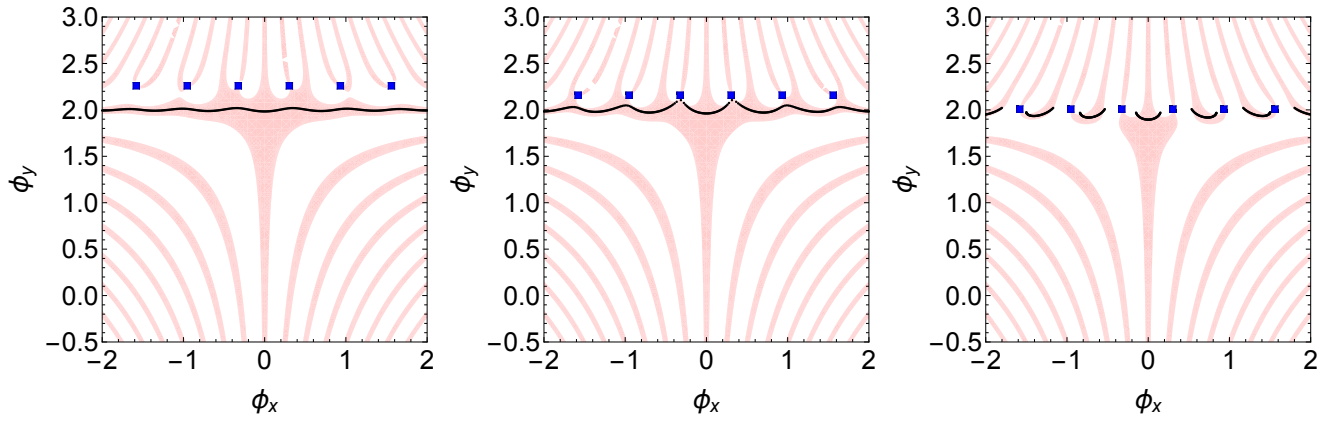
Sometimes, it is also possible to sample from a complex distribution. For example, lets consider

$$S_\mu(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right] . \tag{11}$$

with the path parameter μ. At $\mu = 0$ we can use a decompositions into Gaussians for sampling

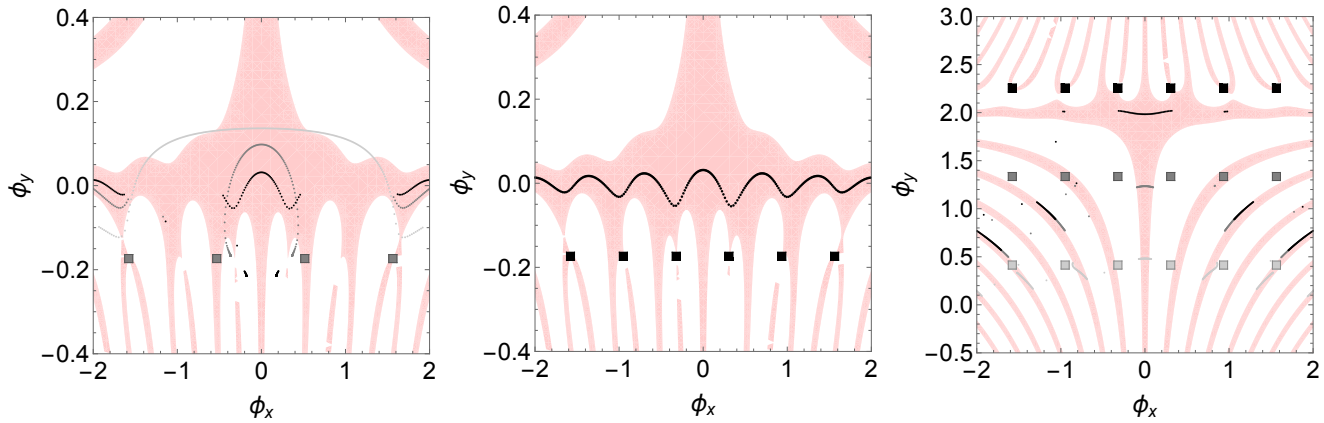
$$p_{\mu=0} = \frac{\mathcal{N}(0,\beta/U) + 2\mathcal{N}(iU,\beta/U) + e^{\beta U}\mathcal{N}(2iU,\beta/U)}{3 + e^{\beta U}} . \tag{12}$$

To sample from the complex Gaussians, we can use a standard complex PIK. If we do that, we obtain a starting



(a) $\mu = 1.75$ (b) $\mu = 1.65$ (c) $\mu = 1.5$

FIG. 1: Final sampling manifold for the single-PIKfold approach discussed in **??** for different densities. The sampling manifold is indicated in *black*, whereas the *blue* dots indicate the poles in the action. The red areas are inserted for orientation and indicate $\mathrm{Im}\, S(\phi) \pmod{2\pi} < \frac{\pi}{4}$ or $\mathrm{Im}\, S(\phi) \pmod{2\pi} > \frac{7\pi}{4}$.

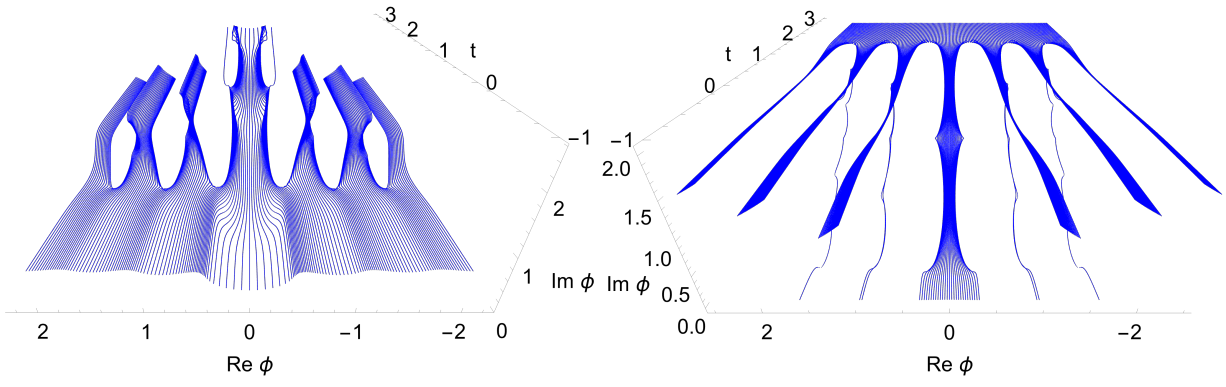


(a) PIKfold at $t = 0.2$ (*light grey*), $t = 0.6$ (*grey*) and $t = 1$ (*black*) for the path **??** with $\mu = -0.675$.

(b) PIKfold at $t = 1$ for the path **??** with $\mu = -0.675$.

(c) PIKfold at $t = 10^{-8}$ (*light grey*), $t = 10^{-4}$ (*grey*) and $t = 1$ (*black*) for the path **??** with $\mu = 2$.

FIG. 2: Time slices of the PIKfold for the different path and parameter choices of **??** and **??** at $\beta = 10$ and $U = 1$. (b) Shows a closed manifold that can be sampled from, whereas (a, c) show a ripping of the PIKfold.



(a) $\mu_0 = 0$ (b) $\mu_0 = 2$

FIG. 3: PIKfold.

manifold $\mathcal{M}_0 = \mathbb{R} + \frac{2}{3}i$. This we can use to evolve our PIK along μ. The PIKfold for $\mu \in [-1, 2]$ is displayed in **??**. While the PIKfold is well defined for $\mu < 0$, we find ripping at $\mu \approx 0.1$. This is shown also in **??**.

V. EXTENSION TO WEIGHTED SUMS

In cases with a particularly severe sign-problem, as e.g. QCD at finite density the simple complex PIK may fail to provide a closed contour/manifold $\mathcal{C}_1$. This is the case if *(i)* such a contour does not exist in the entirety of $\mathbb{C}^{\mathcal{D}}$ at $t = 1$, or if *(ii)* a suitable contour at $t = 1$ cannot be linked to the initial contour at $t = 0$ since the path of the action $S(t)$ is not connecting them. [**FI:** This is actually the case at $\mu = 1/2$ in the example considered below!]

The failure of the complex PIK in these cases is signalled by the contour ripping open [**FI:** (the naive check if a contour fails is to look for large derivatives in the map).] The situation can be remedied by splitting up the distribution as a weighted sum

$$p_t(\phi) = \frac{\sum_i \mathcal{N}_i \, p_{i,t}(\phi)}{\sum_i \mathcal{N}_i} \,, \tag{13}$$

where $\mathcal{N}_i$ is the normalisation of the corresponding p_i.

The amount distributions we chose for the split can be optimised for the corresponding problem.

With this split contributions with different phases can be absorbed into different terms.

A. Example

We are considering the action

$$S_t(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{t\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right] . \tag{14}$$

with the one-site interaction parameter U, inverse temperature β, density μ and RG-time t At $t = 0$ this simply corresponds to a free action with mass β/U, whereas we obtain the *one-site Hubbard model* from [**?**] at $t = 1$.

As an advantage of the one-site model we can disentangle the action and rewrite it as a sum of three Gaussian distributions

$$\begin{aligned} e^{-S_t(\phi)} &= e^{-\frac{\beta}{2U}\phi^2} + 2e^{t\beta\left(\mu+\frac{U}{2}+i\phi\right)-\frac{\beta}{2U}\phi^2} \\ &\quad + e^{2t\beta\left(\mu+\frac{U}{2}+i\phi\right)-\frac{\beta}{2U}\phi^2} \,, \end{aligned} \tag{15}$$

each of these contributions is a (shifted) Gaussian and allows for the analytical computation of a sign-problem free contour using the PIK approach. This already allows for the following observations: In different parameter ranges $\mu, \beta, \ldots$ a different distribution dominates the physics. For example at $\mu + U/2 \ll 0$ the last two terms are suppressed and the contour stays very close to the real axis - on the other hand large μ mainly draw from the last term in **??**.

Instead of using a single complex PIK, we proceed to add three, corresponding to the respective summands in **??**. The first term remains as is: We sample from the real Gaussian. The second and third term are described by an analytically given PIK, which solves the equation

$$\beta\left(\mu + \frac{U}{2} + i\phi\right) + \partial_t n = \partial_\phi\dot{\phi} - \dot{\phi}\left(it\beta - \frac{\beta}{U}\phi\right) , \tag{16}$$

where t is integrated up to $t = 1$ and $t = 2$ for the second and third term respectively. This ODE has a general analytic solution, which is only finite for all ϕ if the normalisation is chosen as

$$\begin{aligned} \partial_t n &= -\beta\left(\mu + \frac{U}{2} - tU\right) , \\ \Rightarrow n_t &= -\beta\left(\mu t + \frac{U}{2}t(1-t)\right) + n_0 \,, \end{aligned} \tag{17}$$

where n_0 is the normalisation of the original Gaussian at $t = 0$. With this the solution to the PIK follows as

$$\dot{\phi} = iU \quad \Rightarrow \quad \phi_t(\varphi) = \varphi + iUt \,, \tag{18}$$

where φ is given by the original real field $\varphi = \phi_0 \in \mathbb{R}$ and

$$\varphi \sim e^{-\frac{\beta}{2U}\varphi^2} . \tag{19}$$

By weighting the total distribution with the adjusted normalisations we can now sample general, holomorphic operators $O(\phi)$ with

$$\langle\hat{O}\rangle = \sum_i \frac{O(\varphi_i) + 2N_1 O(\varphi_i + iU) + N_2 O(\varphi_i + 2iU)}{1 + 2N_1 + N_2} , \tag{20}$$

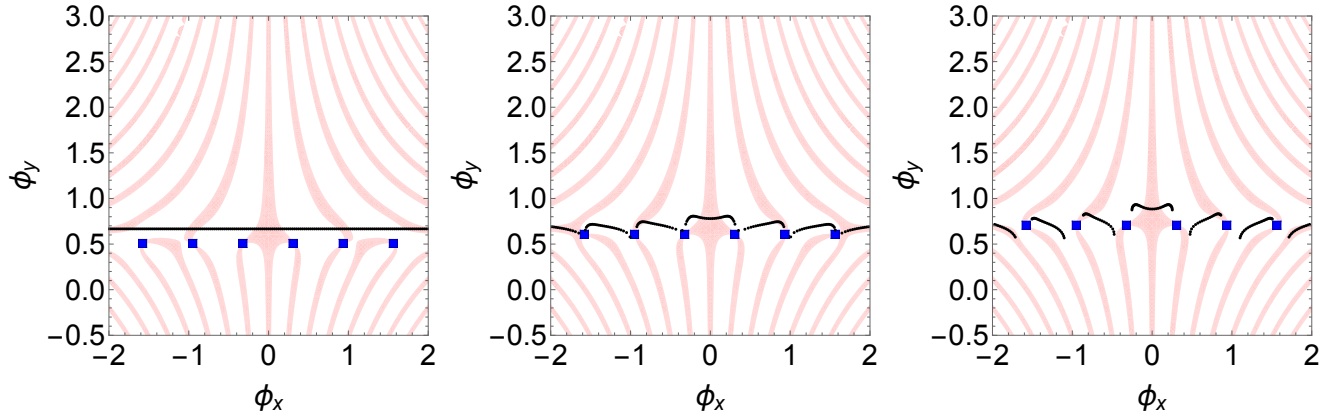
where $N_t = e^{n_t}$ and we sum over Gaussian samples φ_i. To verify the calculation we sample the density operator, which is given by

$$n = -i\langle\phi\rangle . \tag{21}$$

The μ-dependence of the density at $\beta = 10$ and $U = 1$ is displayed in **??**. We also show the solution of a direct evaluation of the integral as a comparison.

VI. RESOLVING THE PIK PAIR

Here we discuss constructive solutions for the determination of the PIKfold, or the PIKsubfolds **??**, specifically in the case of higher-dimensional fermionic theories. To begin with, we note that the distribution of many theories resembles closely weighted sums such as **??**. This certainly holds true for the one-site Hubbard model but extends beyond this very restrictive example. Moreover,



(a) $\mu = 0.0$ (b) $\mu = 0.1$ (c) $\mu = 0.2$

FIG. 4: Sampling manifold for the single-PIKfold approach discussed in **??** for different densities. The sampling manifold is indicated in *black*, whereas the *blue* dots indicate the poles in the action. The red areas are inserted for orientation and indicate $\mathrm{Im}\, S(\phi) \pmod{2\pi} < \frac{\pi}{4}$ or $\mathrm{Im}\, S(\phi) \pmod{2\pi} > \frac{7\pi}{4}$.

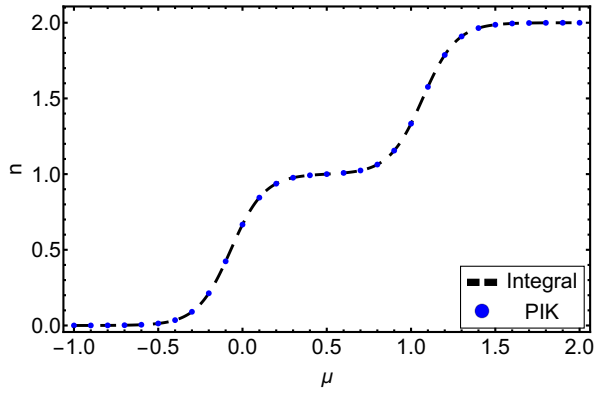


FIG. 5: Density **??** as a function of chemical potential μ for couplings $U = 1$ and $\beta = 10$.

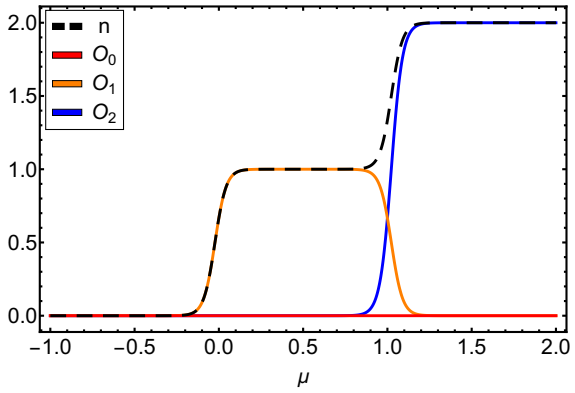


FIG. 6: Density **??** as a function of chemical potential μ for couplings $U = 1$ and $\beta = 30$. We also depict the contributions from the different PIKs **??**.

many multimodal distributions can be expanded about a weighted sum of normal distributions that are peaked at the peaks of the full distributions.

Moreover, in terms of PIKfolds these weighted sums can be collected into equivalence classes: two distributions $p(\phi)$ and $q(\phi)$ fall into the same equivalence class if they can be connected with a PIKfold $\mathcal{P} = \{\mathcal{M}_t\}$,

$$p \simeq q : \quad \exists\, (p_t\,, \mathcal{P}) \quad \text{with} \quad p_0 = p\,, \quad p_1 = q\,. \tag{22}$$

Note that the equivalence **??** may be hard to prove as we can start with any sampling manifold $\mathcal{M}_0$. In the cases discussed below we start with the original sampling manifold of the underlying field theory, in our example theory, the one-site Hubbard model this is $\mathcal{M}_0 = \mathbb{R}$.

The key idea of the constructive solution of the sampling problem is the following.

(1) Finding p_t, weighted sum **??** or similar, as with $p_1 \simeq e^{-S}$.

(2) Sampling from p_t.

Both steps are sufficiently generic to allow for different solving strategies best-suited for specific problems. Below we shall discuss several ones for (1) and a tailor made one for (2).

A. Resolving the complex structure

In this Section we work out and test a two step approach to the PIK-sampling of complex distributions in the presence of ripping. In the first step we resolve the ripping with the PIK-pair

$$(\bar{S}_t\,, \bar{\mathcal{M}}_t)\,, \tag{23}$$

without ripping for all $t \in [0, 1]$ and $(\bar{\rho}_0\,, \bar{\mathcal{M}}_0)$ being real for the sake of simplicity. Other choices of the initial

sampling pair are possible and do not modify the procedure. Evidently such a choice excludes $\bar{S}_1 = S$ if the latter does not allow a PIKfold. Accordingly we have

$$\bar{S}_t = S'_t + \mathcal{T}_t\,, \qquad S'_0 \equiv 0\,, \quad S'_1 = S\,. \tag{24}$$

with $\bar{p}_0 = \exp\{-\mathcal{T}_0\}$ is the initial simple distribution and $\mathcal{T}_t$ prevents the ripping of the manifold $\mathcal{M}_t$. Specifically, T_1 arranges for $\mathrm{Re}(\bar{p}_1) \neq 0$ on $\mathcal{M}_1$. We know from [**?**] that the set of $\mathcal{T}_1$ can be computed from S alone: all admissible $\mathcal{T}_t$ can be inferred from the condition $\mathrm{Re}(\bar{p}_1) \neq 0$ on $\bar{\mathcal{M}}_1$. However, in a high-dimensional data space this entails the search of critical points in a high dimensional space, and in general the computational complexity of this task rivals that of solving sign problems. Moreover, we also need a trajectory $\mathcal{T}_t$ which prevents the ripping of $\bar{\mathcal{M}}_t$ for all t.

The above intricacies suggest a flowing approach to this problem that works with a qualitative resolution of the PIK $\dot{\phi}$: Core of the procedure is the choice of the sampling pair **??** or rather the 'trial' or prior part $\mathcal{T}_t$: First of all, it has to be chosen such that it allows an analytic access, or, more generally, the possibility to sample on $(\mathcal{T}_1, \bar{\mathcal{M}}_0)$ directly or via a PIK-architecture. A choice tailor made for the present task is given by

$$\mathcal{T}_t = \log \sum_{i\geq 0}^{N_t} c_{i,t}\rho_{i,t}(\phi)\,, \qquad c_{0,0} = 1\,, \quad \rho_{0,0} = \bar{\rho}_0\,, \tag{25a}$$

[**JMP:** normalisation has to be included, I'm not sure concerning the 'best' parametrisation] and Gaußian distributions

$$\rho_i(\phi) = e^{-\frac{1}{2}(\phi_n - \phi_n^{(i)})\mathcal{C}^{(i)}_{nm}(\phi_m - \phi_m^{(i)})}\,, \tag{25b}$$

with the t-dependent center $\phi^{(i)}$ of the distribution ρ_i and the t-dependent covariance $\mathcal{C}C$. In general, all these parameters are complex and vanish at $t = 0$,

$$\phi^{(i)}_{m,t}\,, \mathcal{C}^{(i)}_{nm,t} \in \mathbb{C}\,, \qquad \phi^{(i)}_{m,0}\,, \mathcal{C}^{(i)}_{nm,0} = 0\,. \tag{25c}$$

The t-dependence of the parameters in **??** is chosen with the following constraint, or similar ones. The solution of the PIK $\dot{\phi}_t$ with $\mathcal{T}_t = 0$ for $t < t_{\mathrm{rip}}$ with t_{rip} will experience growing ratios

$$\left|\frac{\mathrm{Im}\left[\bar{p}_t(\phi^{(i)})\right]}{\mathrm{Re}\left[\bar{p}_t(\phi^{(i)})\right]}\right| \to \infty\,, \quad \text{for} \quad t \to t_{\mathrm{rip}}\,. \tag{26}$$

Here, t_{rip} is the flow time where the ripping occurs and $(\phi^{(i)})_t$ are the peak positions of these ratios. Importantly for us, these emerging singular points are the result of the resolution of the Wegner equation for $\dot{\phi}$ and do not have to be searched for. Now we choose $\mathcal{T}_t$ such that

$$\left|\frac{\mathrm{Im}\left[\bar{p}_t(\phi)\right]}{\mathrm{Re}\left[\bar{p}_t(\phi)\right]}\right| < M\,, \tag{27}$$

with a suitable chosen maximal value M. Not that this flow procedure provides us with the number and location $\phi^{(i)}$ of the ripping points as well as constraints for the matrix elements of the covariance, $\mathcal{C}_{mn}$. Naturally, the latter parameters are not uniquely determined while $\phi^{(i)}$ is uniquely defined as the peak position [**JMP:** Also this is not completely true as we may shift the peak position with $\mathcal{T}_t$ but who cares].

[**JMP:** Not fully checked from now on]

Importantly, such a procedure does not involve a quantitative resolution of the PIK $\dot{\phi}$ but works with a qualitative numerically cheap one as long as the ripping points are not 'fractal-type'.

After the $\mathcal{T}_t$ is determined within the above first step, we define the final sampling procedure within a second step. To that end we define the final PIK-architecture as the pair

$$(S_t\,, \qquad \mathcal{M}_0)\,, \quad S_t = \bar{S}_t - \mathcal{T}_1 = S'_t + [\mathcal{T}_t - \mathcal{T}_1]\,, \tag{28a}$$

and e.g. the real sampling pair

$$(S_0\,, \mathcal{M}_0)\,, \qquad p_0(\phi_0)d\mu_0 \in \mathbb{R}\,. \tag{28b}$$

The latter condition **??** is one of the possible choice. Evidently, the sampling of

. (29)

B. The maximal PIKfold

Here we recall the reality condition eq.(50) in [**?**]. There it was formulated for PIKfolds that can be parametrised with $\sigma = \phi_R$. More generally, we have

$$\frac{d\phi_I}{d\sigma}\mathrm{Re}\left[p_t(\phi)\right] = -\frac{d\phi_R}{d\sigma}\mathrm{Im}\left[p_t(\phi)\right]\,. \tag{30}$$

Note that **??** does not exhibit the cuts which are present in the likelihood S and in the absence of cuts in p_t, **??** only exhibits poles. Bearing this in mind we may rewrite **??** in terms of the likelihood

$$\phi'_I\,\cos\left[S_{I,t}(\phi)\right] = \phi'_R \sin\left[S_{I,t}(\phi)\right]\,, \quad \phi'(\sigma) = \frac{d\phi(\sigma)}{d\sigma}\,, \tag{31}$$

with $p_t = \exp\{-S_t\}$ with $S_t = S_R + \mathrm{i}\,S_I$. Evidently, for a real action S_t, all manifolds in a PIKfold agree, $\mathcal{M}_t = \mathcal{M}_0$ (in the absence of stretching singularities). Moreover, for configurations ϕ_{sing} with $\mathrm{Re}[p_t(\phi)] = 0$, the vector field (ϕ'_R, ϕ'_I) is perpendicular to the real manifold $\{\Phi_R\} = \mathbb{R}^{\mathcal{D}}$. Note that for the choice $\sigma = \phi_R$ this leads to a ripping of the parametrisation, not necessarily the ripping of the PIKfold. However, $\dot{\phi}(\phi)$ also exhibits a singularity in this point which begs the question whether the parametrisation of $\dot{\phi}$ in terms of ϕ is a suitable one in such a case. More practically we have to sort out whether **??** can be used to defined $\mathcal{M}_1$ with $\sigma = -\infty$ with the initial condition $(\phi_R = \infty, \phi_I)$ to $\phi_R = \infty$.

ACKNOWLEDGMENTS

We thank XXX for discussions. This work is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy EXC 2181/1 - 390900948 (the Heidelberg STRUCTURES Excellence Cluster), the Collaborative Research Centre SFB 1225 - 273811115 (ISOQUANT) and the Walter Benjamin Programme - 574031240. This work is also supported by the KAKENHI grant "Foundation of Machine Learning Physics."

Appendix A: PIKs & Lefschetz thimbles approach

In this section we discuss parallels and crucial differences between the complex PIKs and the Lefschetz thimble approach. By applying the *target action* strategy to computing thimbles, we find a semi-analytic strategy to compute the thimble solutions and to sample from them.

We begin with a brief introduction to Lefschetz-thimbles in **??**. This is followed by the construction of Lefschetz-thimbles from a complex PIK in **??**.

1. A brief repetition of Lefschetz-thimbles

Similar to the complex PIK architecture, Lefschetz thimbles (LT) use the strategy of contour deformation to attack the sign problem. In distinction to the former, the LT-contour is made up of the so-called thimbles $\mathcal{J}_\sigma$ on which the phase of the integrand is constant. The thimbles are computed as follows: Firstly, every thimble is attached to a critical point ϕ_σ of the action given by

$$\frac{\partial S}{\partial \phi}(\phi_\sigma) = 0\,. \tag{A1}$$

Then, the thimble $\mathcal{J}_\sigma$ is then defined as the set of points obtained from the downward gradient flow

$$\frac{d\phi}{dt} = \overline{\left(\frac{\partial S}{\partial \phi}\right)}\,, \tag{A2}$$

with the boundary condition $\phi_{t\to-\infty} = \phi_\sigma$. By considering

$$\frac{d}{dt} S(\phi)\bigg|_{\mathcal{J}_\sigma} = \frac{\partial S}{\partial \phi}\overline{\left(\frac{\partial S}{\partial \phi}\right)} = \left|\frac{\partial S}{\partial \phi}\right|^2\,, \tag{A3}$$

we find that along **??**, the real part of the action increases monotonically, while its imaginary part remains constant. The former property implies that the critical points are the leading contribution to the sampling procedure.

The sampling contour is a linear combination of all thimbles,

$$\mathcal{C} = \sum_\sigma n_\sigma, \mathcal{J}_\sigma\,, \tag{A4}$$

where the coefficients $n_\sigma \in \mathbb{Z}$ are intersection numbers of the corresponding anti-thimbles with the real axis. The anti-thimbles are the result of the flow **??** with the boundary condition $\phi_{t\to+\infty} = \phi_\sigma$. Finally, the partition function can be written as a sum of integrals over the contributing thimbles

$$Z = \sum_\sigma n_\sigma\, e^{-i\,\mathrm{Im}S(\phi_\sigma)} \int_{\mathcal{J}_\sigma} d\mu_\sigma\, e^{-\mathrm{Re}S(\phi)}\,. \tag{A5}$$

In general $\mathcal{J}_\sigma$ is a curved manifold/contour in a complex space $\mathbb{C}^{\mathcal{D}}$, with a complex measure $d\mu_\sigma$. Hence the integrals over single thimbles $\mathcal{J}_\sigma$ have to be rewritten in terms of a real parametrisation. For the single-site example we can choose $\phi(s)$ with $s \in \mathbb{R}$. Then the integral over a single thimble reads

$$Z_\sigma = \int ds \det\left(\frac{\partial \phi}{\partial s}\right) e^{-S(\phi(s))}\,. \tag{A6}$$

The constant phase factor in the action can be factored out, however the Jacobi-determinant $\det\left(\frac{\partial\phi}{\partial s}\right)$ is not real in general, which leaves one with a residual phase

$$\theta(s) = \arg\det\left(\frac{\partial \phi}{\partial s}\right) - \mathrm{Im}\, S(\phi_\sigma)\,. \tag{A7}$$

Expectation values of observables are then computed by re-weighting with the residual phase

$$\langle \mathcal{O}\rangle = \frac{\int ds\, P(s)\, e^{i\theta(s)} \mathcal{O}(\phi(s))}{\int ds\, P(s)\, e^{i\theta(s)}}\,, \tag{A8a}$$

with the real weight

$$P(s) = \left|\det\left(\frac{\partial \phi}{\partial s}\right)\right| e^{-\mathrm{Re}\, S(\phi(s))}\,. \tag{A8b}$$

In conclusion, the LT approach uses a decomposition of the contour into different thimbles. While the phase on a specific thimble is constant and sampling can be performed after a re-weighting, the phases of two different thimbles are generically different, which can lead to large cancellations and a residual sign problem. Moreover, for many systems the total number of thimbles is very large, reintroducing the sign problem through the back door.

2. Thimbles through the PIK-lens

The complex PIK architecture builds on the Wegner equation which considers general reparametrisations of the theory. As such, it is very flexible in its choice of the initial sampling manifold, and the path with which we deform the action. For the aim of the present section, i.e. discussing the embedding of the Lefschetz thimble approach in the PIK, we restrict ourselves to real initial sampling manifolds

$$(p_0(\phi)\,,\, \mathcal{M}_0)\,, \qquad \mathcal{M}_0 = \mathbb{R}\,. \tag{A9}$$

This choice is made for simplicity, and allows to use the initial real variable $\phi_0 \in \mathbb{R}$ as parametrisation for the complex sampling manifold or complex integration contour $\phi_1(\phi_0) \in \mathcal{M}_1$.

As already noted previously, the thimble approach and the complex PIK follow a similar strategy by deforming the contour. However, we find that there are two key differences:

(a) There is only one final sampling manifold for the complex PIK, in distinction to the thimble approach, where there can be as many thimbles as there are critical points **??**.

(b) The thimble aims for a constant phase on a curved manifold, which leaves one with the residual phase **??** containing the Jacobi-determinant of the real parametrisation. In contrast, the parametrisation of the complex contour is already part of the weight-preserving property of the complex PIK, effectively achieving $\theta(\phi_0) = 0$ on the entire PIKfold $\mathcal{M}_t$.

To illustrate (b), we recall the weight preserving property of the PIK, which reads

$$\int_{\mathcal{M}_t} d\mu_t \, e^{-S_t(\phi_t)} = \int_{\mathcal{M}_0} d\mu_0 \det\left[\frac{\partial \phi_t}{\partial \phi_0}\right] \, e^{-S_t\left(\phi_t(\phi_0)\right)} \equiv \int_{\mathcal{M}_0} d\mu_0 \, e^{-S_0(\phi_0)} \,, \tag{A10}$$

and compare it to **??**: The important difference is, that following the second equality, the product of the Jacobi-determinant and the exponential of the action have a constant phase. Even better, they are completely real, whereas the LT constructs only an action with a constant phase but leaves the parametrisation of the manifold out. The inclusion of the Jacobian into the definition of the PIK is the crucial difference between both approaches. In fact, in systems where $\det\left[\frac{\partial \phi_t}{\partial \phi_0}\right] = \text{const.}$ the complex PIK and the LT describe the same manifold.

a. *Constructing Lefschetz-thimbles from the complex PIK*

In this section we discuss how LT's can be computed within the complex PIK framework. To this aim, we go back to the more general considerations about the *physics-informed* pair (PIRG-pair)

$$\left(S_t(\phi), \dot{\phi}(\phi)\right) \,. \tag{A11}$$

For the complex PIK applications in [**?**] we have defined $S_t(\phi)$ with a unique path and proceeded with computing the resulting $\dot{\phi}$ from the Wegner equation **??**. However, as seen in applications within the fRG [**?** **?** **?** **?**] it can make sense to use the reparametrisation freedom in the Wegner equation differently, i.e. by implementing a different, additional constraint instead of completely fixing the path of the action.

In the present case, we want to use this freedom to implement a generative architecture that samples on the Lefschetz thimble. To this aim we define a path $\tilde{S}_t$ for the action, which starts with an easy distribution on the real axis and evolves towards our target distribution on the thimble

$$\left(e^{-\tilde{S}_0(\phi)} \,, \mathbb{R}^{\mathcal{D}}\right) \longrightarrow \left(e^{-\tilde{S}_1(\phi)} \,, \mathcal{J}_\sigma\right) \,. \tag{A12}$$

Note our choice of notation $\tilde{S}_t$: This path does not correspond to the action in the PIRG-pair, it is simply a tool to construct a kernel $\dot{\phi}$ which fulfils

$$0 \stackrel{!}{=} \partial_t \tilde{S}_t(\phi_t) = \partial_t \tilde{S}_t(\phi_t) + \partial_t \phi_t \, \partial_{\phi_t} \tilde{S}_t(\phi_t) \,, \quad \dot{\phi} := \partial_t \phi_t \,. \tag{A13}$$

Since the initial $\tilde{S}_0$ is real, **??** ensures that all $\tilde{S}_t$ and importantly $\tilde{S}_1$ on the thimble are also real. Note that this allows for a complex phase, as

$$\tilde{S}_t = \log\left(\tilde{\mathcal{N}}_t\right) + \tilde{\tilde{S}}_t \,, \tag{A14}$$

contains a normalisation term.

The normalisation term in **??** is uniquely determined by requiring that the thimble remains attached to an evolving critical point **??**

$$\dot{\phi}(\phi_{\sigma,t}, \tilde{\mathcal{N}}_t) \stackrel{!}{=} \partial_t \phi_{\sigma,t} = -\frac{\partial_t \tilde{S}_t^{(1)}(\phi_{\sigma,t})}{\tilde{S}_t^{(2)}(\phi_{\sigma,t})} \,, \tag{A15}$$

where the second equality is derived by requiring that $\partial_t \tilde{S}^{(1)}(\phi)|_{\phi=\phi_\sigma} = 0$, which simply says that the critical point is defined by its vanishing first derivative at all times.

Together **??** and **??** yield a simple algebraic equation to compute the kernel. If we restrict ourselves to paths $\tilde{S}_t$ where the total number of critical points is preserved, each initial critical point allows to compute a different final thimble.

In distinction to the standard complex PIK procedure, **??** does not allow for straightforward sampling. In fact, evaluating the operators directly using **??**, simply evaluates the distribution **??** containing the Jacobian of the curved contour. In this sense, the kernel defined in **??** describes a weight-preserving complex PIKfold for the target distribution

$$S_t(\phi) = \tilde{S}_t(\phi) + \operatorname{Tr} \log\left[\frac{\partial \phi_t}{\partial \phi_0}(\phi_0)\right]\Big|_{\phi_0 = \phi_0(\phi_t)} \,, \tag{A16}$$

which one can also compute by solving the Wegner equation **??** using the kernel **??**. This computation is of course much more difficult, as we have to solve a partial differential equation.

To obtain observables of the action $\tilde{S}_t(\phi)$ one can simply resort so re-weighting in the standard way **??**. For the generated samples this corresponds to summing

$$\langle \mathcal{O}(\phi)\rangle_{\tilde{p}_t} = \frac{\sum_n \mathcal{O}(\phi_t(\phi_0)) \det\left[\frac{\partial \phi_t}{\partial \phi_0}(\phi_0)\right]}{\sum_n \det\left[\frac{\partial \phi_t}{\partial \phi_0}(\phi_0)\right]} \,. \tag{A17}$$

Note, that the quality of results is expected to be worse for steeper slopes in the Jacobian. On the plus-side we have semi-analytics access to the thimble.

Appendix B: Remnants

1. Complex sampling and the sign problem

PIK-architectures [**?**] have been mainly developed for the efficient sampling with normalised distributions of lattice field theories. In the present work we consider a scalar field theory with a real scalar field and a generic complex distribution for illustrative purposes. We define

$$p(\varphi) = e^{-S(\varphi)} \in \mathbb{C} \,, \qquad S(\varphi) = \hat{S}(\varphi) + \log \mathcal{N} \,. \tag{B1a}$$

Here, $S(\varphi) \in \mathbb{C}$ is the action of the complex distribution $p(\varphi) \in \mathbb{C}$, with $\hat{S}(\varphi = 0) = 0$. The constant term $\log \mathcal{N} \in \mathbb{C}$ is nothing but the normalisation of the distribution $p(\varphi)$ which guarantees

$$Z = \int\limits_{\mathcal{M}} d\varphi \, p(\varphi) = 1 \,, \qquad \int\limits_{\mathcal{M}} d\varphi = \prod_i \int\limits_{\mathbb{R}} d\varphi_i \,. \tag{B1b}$$

where i labels the data points. In our lattice example, the *real* field φ is defined on a finite (hyper-)cubic lattice $\mathcal{D}$ with

$$\varphi : \mathcal{D} \to \mathcal{M} = \mathbb{R}^{\mathcal{D}} \,, \qquad \mathcal{D} = [0, 1, ..., N_{\max}]^d \,. \tag{B1c}$$

We will not specify the action $\hat{S}(\varphi)$; typically it includes a kinetic term (hopping term) and an on-site self-interaction (potential term). For such an example of a real scalar lattice field theory we refer the reader to [**?**]. In the general setup, $\mathcal{D}$ labels a continuous or discrete data set with data φ, and the continuous or discrete sampling manifold $\mathcal{M}$ is generalised likewise. All these setups are defined by *sampling pairs* $(p(\varphi)\,, \mathcal{M})$ of the probability distribution and the sampling manifold.

In general, sampling with a complex distribution $p(\varphi) \in \mathbb{C}$ in **??** poses a numerical challenge, and for highly oscillatory distributions the sampling task may exhibit a sign problem. A rather generic and seemingly simple resolution of such a sign problem in our example **??** is provided by the deformation of the integration manifold $\mathcal{M}$ with a highly oscillatory distribution $p(\varphi)$ into an integration manifold $\mathcal{M}_1 \subset \mathbb{C}^{\mathcal{D}}$, on which $p(\varphi)$ exhibits no sign problem.

Assuming that this deformation step has been performed, it leaves us with a secondary problem of efficiently sampling from the given distribution $p(\varphi)$. This secondary problem can be remedied by mapping $p(\varphi)$ to a simple distribution such as a normal distribution $p_0(\phi)$ where $\phi \in \mathcal{M}_0 \subset \mathbb{C}^{\mathcal{D}}$. While normal distributions with the sampling manifold $\mathcal{M}_0 = \mathcal{M}$ are standard choices, the set of viable $(p_0, \mathcal{M}_0)$ is not restricted to them. The only requirement is efficient sampling while keeping all symmetries.

2. Complex sampling with physics-informed kernels

In the present work we use the PIK-architecture [**?**] to resolve the combined sign problem and efficient sampling tasks discussed above: in general, the physics-informed kernel (PIK) provides a map from some sampling pair $(p_0(\phi)\,, \mathcal{M}_0)$ to that under investigation,

$$(p_0(\phi)\,, \mathcal{M}_0) \longrightarrow (p_1(\phi)\,, \mathcal{M}_1) \,, \qquad p_1(\phi) = p(\phi) \,. \tag{B2}$$

Consequently, the PIK-architecture allows for an efficient sampling of the pair $(p(\phi)\,, \mathcal{M}_1)$, if the original pair can be sampled efficiently. This solves the given sampling task if the sampling on $\mathcal{M}_1$ is equivalent to that on the original sampling manifold $\mathcal{M}$,

$$\int\limits_{\mathcal{M}_1} d\mu_1 \, p(\phi) \, \mathcal{O}(\phi) \stackrel{!}{=} \int\limits_{\mathcal{M}} d\mu \, p(\varphi) \, \mathcal{O}(\varphi) \,, \tag{B3}$$

for general operators $\mathcal{O}(\phi)$. Here, $d\mu_1(\phi)$ and $d\mu(\varphi)$ are the volume elements of $\mathcal{M}_1$ and $\mathcal{M}$, embedded in the complexification $\mathcal{C}$ of $\mathcal{M}$ and φ denotes the variable in the original complex sampling problem. **??** is satisfied if $\mathcal{M}_1$ can smoothly be deformed into $\mathcal{M}$ without swiping over singular points in the distribution $p(\phi)$ and the operators $\mathcal{O}(\phi)$.

In our lattice example the sampling manifold is given by $\mathcal{M} = \mathbb{R}^{\mathcal{D}}$ and its complexification is given by $\mathcal{C} = \mathbb{C}^{\mathcal{D}}$. For the sake of simplicity we will consider this explicit example from now on without loss of generality.

The key property of the PIK-map in **??** is that it is weight-preserving. This entails in particular, that the sampling with $(p_1(\phi)\,, \mathcal{M}_1)$ is real, *if* the sampling pair $(p_0(\phi)\,, \mathcal{M}_0)$ is real. More generally, if the, potentially complex, sampling pair $(p_0(\phi)\,, \mathcal{M}_0)$ can be sampled efficiently and has no sign problem, it induces an efficient and sign- and overlap-problem free sampling for the sampling pair $(p_1(\phi)\,, \mathcal{M}_1)$.

We proceed by discussing the computation of the physics-informed kernel (PIK), that is the differential map $\dot{\phi}(\phi)$. In the PIK-architecture [**?**] the local deformation of the sampling manifold and the distribution is governed by the Wegner equation [**?**],

$$\frac{d\, p_t(\phi)}{dt} = -\frac{\partial}{\partial \phi_i}\Big[\dot{\phi}_{t,i}(\phi)\, p_t(\phi)\Big] \,, \qquad t \in [0, 1] \,, \tag{B4}$$

with the corresponding PIK $\dot\phi_t(\phi)$ of the deformation, and the RG-time (renormalisation group time) t. The index $i \in \mathcal{D}$ in **??** is summed over the data points. **??** depends on a t-dependent normalised distribution

$$p_t(\phi) = e^{-S_t(\phi)}\,, \tag{B5}$$

with $S_t = \hat S_t + \log \mathcal{N}_t$. Importantly, $\hat S_t$ is chosen analytically to interpolate between $p_0(\phi)$ and $p_1(\phi)$. In our lattice example ϕ is the complex field variable on the lattice $\mathcal{D}$,

$$\phi : \mathcal{D} \to \mathbb{C}^{\mathcal{D}}\,. \tag{B6}$$

In general, $\mathcal{D}$ is the set of data points and ϕ takes values in the complexification of the data range.

The complexification of ϕ has two sources. The first rather trivial one is the possible choice of a complex initial manifold $\mathcal{M}_0 \subset \mathbb{C}^{\mathcal{D}}$. The second source is the deformation $\partial_t\mathcal{M}_t$ of the manifold $\mathcal{M}_t$ due to the complex transformation $\dot\phi_t$ itself. With **??**, we rewrite **??** as a differential equation for the physics-informed RG-pair (PIRG pair) $(S_t(\phi)\,,\,\dot\phi)$,

$$\frac{d\,S_t(\phi)}{dt} + \dot\phi_t(\phi)\,\frac{\partial S_t(\phi)}{\partial\phi} = \frac{\partial\dot\phi_t(\phi)}{\partial\phi}\,, \tag{B7}$$

where we have suppressed the indices $i \in \mathcal{D}$ for the sake of readability. For complex $dS_t(\phi)/dt$ and/or $\partial S_t(\phi)/\partial\phi$, the map $\dot\phi(\phi)$ is complex and so is the differential manifold $\partial_t\mathcal{M}_t$. Note that for smooth $\dot\phi(\phi)$, the manifold $\partial_t\mathcal{M}_t \subset \mathbb{C}^{\mathcal{D}}$ is a closed or open hyper-surface, an example with one complex dimension and further explanations can be found in the supplement.

Accordingly, **????** comprise a general differential reparametrisation of the path integral with a smooth deformation of the integration manifold with the PIK $\dot\phi(\phi) \in \mathbb{C}^{\mathcal{D}}$, if the latter is invertible and non-singular.

Now we set up the complex PIK-architecture for the sign-problem free sampling task: With **????** we map the sampling with $p(\varphi)$ on $\mathcal{M}$ with a sign problem to the efficient sampling with $p_0(\phi)$ on the sign-problem free manifold $\mathcal{M}_0$. We find for all t,

$$\int\limits_{\mathcal{M}_t} d\mu_t\, p_t(\phi)\,\mathcal{O}(\phi) = \int\limits_{\mathcal{M}_0} d\mu_0\, p_0(\phi)\,\mathcal{O}\big(\phi_t(\phi)\big)\,, \tag{B8a}$$

where

$$\phi_t(\phi) = \phi + \int_0^t ds\,\dot\phi_s(\phi_s)\,, \qquad \varphi(\phi) = \phi_1(\phi)\,, \tag{B8b}$$

and $d\mu_t$ denotes the volume element of the corresponding manifold $\mathcal{M}_t$. **??** with **??**, evaluated for $t = 1$, leads to the aimed for equivalence

$$\int\limits_{\mathcal{M}} d\mu\, p(\varphi)\,\mathcal{O}(\varphi) = \int\limits_{\mathcal{M}_0} d\mu_0\, p_0(\phi)\,\mathcal{O}\big(\varphi(\phi)\big)\,. \tag{B9}$$

The existence of the global map **??** was discussed in detail in [**?**] for real actions S_t and the existence criteria apply to complex S_t without modification. The proof of **??** as well as further details can be found in the supplement. Here we only emphasise that the physics or more generally the complete information of the underlying distribution $p(\varphi)$ is carried by the map $\varphi(\phi)$ alone. The analytic access to this map, and the factorisation of the global computing task into the independent ones of computing $\dot\phi_t(\phi)$ for all $t \in [0,1]$ are amongst the advantageous properties of the PIK-architecture, see again [**?**].

It is left to prove that the right-hand side of **??** is sign-problem free. For that purpose we show, that the sampling with $p_t(\phi)$ on $\mathcal{M}_t$ has no sign and overlap problems, if the sampling with $(p_0(\phi)\,,\,\mathcal{M}_0)$ has no sign problem. This important property can be deduced from the *weight-preserving* property of the PIK-architecture. For that purpose we shall prove that the flow of the 'statistical' weight of general intervals $\mathcal{T}_t$ vanishes, that is

$$\frac{d\,P_t(\mathcal{T}_t)}{dt} = 0\,, \qquad P_t(\mathcal{T}_t) = \int_{\mathcal{T}_t} d\mu_t\, p_t(\phi)\,. \tag{B10}$$

Here, $\mathcal{T}_t$ is the image of a given interval $\mathcal{T}_0$ with the map $\phi_t(\phi)$ in **??**. A direct proof of **??** with **??** is provided in the supplement. Here, we simply consider the integral of **??** from the initial RG-time to t, which is $P_t(\mathcal{T}_t) = P_0(\mathcal{T}_0)$. This identity can be derived from **??** by using the characteristic function $\chi_{\mathcal{T}_t}$ of the sub-manifold $\mathcal{T}_t$ as the operator $\mathcal{O}$. Then, $\chi_{\mathcal{T}_t}(\phi_t(\phi)) = \chi_{\mathcal{T}_0}(\phi)$ is the characteristic function of $\mathcal{T}_0$ on $\mathcal{M}_0$. Hence, inserting this operator in **??** leads us to

$$\int\limits_{\mathcal{T}_t} d\mu_t\, p_t(\phi) = \int\limits_{\mathcal{T}_0} d\mu_0\, p_0(\phi)\,, \tag{B11}$$

and the t-derivative of **??** leads us to **??**. If we use infinitesimal sub-manifolds $\mathcal{T}_t$, we readily find that the differential distribution or weight $d\mu_t\, p_t(\phi)$ is equivalent to $d\mu_0\, p_0(\phi)$. The differential equivalence is elucidated with the pull-back of the left-hand side to the pre-image $\mathcal{T}_0$, to wit,

$$\int\limits_{\mathcal{T}_0} d\mu_0 \det\left[\frac{\partial\phi_t}{\partial\phi_0}\right] e^{-S_t\left(\phi_t(\phi_0)\right)} \equiv \int\limits_{\mathcal{T}_0} d\mu_0\, e^{-S_0(\phi_0)}\,. \tag{B12}$$

As **??** holds true for general sub-manifolds $\mathcal{T}_0$, it has to hold true for the integrands, and we arrive at

$$S_t\big(\phi_t(\phi_0)\big) - \mathrm{Tr}\log\left[\frac{\partial\phi_t}{\partial\phi_0}(\phi_0)\right] = S_0(\phi_0)\,. \tag{B13}$$

In short, **??** or **??** entail that the map $\dot\phi$ is weight-preserving by construction and hence no overlap problem and sign problem is generated by the flow with $\dot\phi$. Hence, if the sampling pair at $t = 0$ is chosen sign-problem free, so are all sampling pairs. In particular, if the sampling

pair at $t = 0$ is real, all sampling pairs are real, and specifically $(p_1(\phi)\,, \mathcal{M}_1)$. We call the set of weight-preserving manifolds

$$\mathcal{P} = \mathcal{P}_1\,, \qquad \mathcal{P}_t = \{\mathcal{M}_s \,|\, s \in [0,t]\}\,, \tag{B14}$$

endowed with the holomorphic map $\varphi(\phi)$ on $\mathcal{P}$, the *PIKfold* of the set of flowing distributions $\{p_t\}$. The sets $\mathcal{P}_t$ are sub-PIKfolds. More details are provided in the supplement. Sign-problem free PIK-architectures are obtained by construction for a sign-problem free sampling pair $(p_0, \mathcal{M}_0)$.

[**JMP:** end complex PIK paper [**?**], new from now on]

[] F. Ihssen, R. Kapust, and J. M. Pawlowski, (2026), arXiv:2603.03159 [hep-lat].

[] Y. Tanizaki, Y. Hidaka, and T. Hayata, New J. Phys. **18**, 033002 (2016), arXiv:1509.07146 [hep-th].

[] F. Ihssen, R. Kapust, and J. M. Pawlowski, (2025), arXiv:2510.26678 [hep-lat].

[] F. J. Wegner, Journal of Physics C: Solid State Physics **7**, 2098 (1974).

[] F. Ihssen and J. M. Pawlowski, Annals Phys. **481**, 170177 (2025), arXiv:2409.13679 [hep-th].